Input-Regulated Remote Sensing Counting With Region Understanding

Shengqin Jiang, Haojian Long, Fengna Cheng, Yuankai Qi, Xiaobo Lu, and Qingshan Liu, *Senior Member, IEEE*

***Abstract*—Remote sensing counting aims to automatically estimate the number of objects of interest from high-resolution aerial or satellite imagery, providing critical decision-making support in areas such as urban planning, traffic monitoring, and disaster response. While most existing methods leverage pretrained models to enhance feature generalization, their performance is often hindered by the severe scarcity of annotated remote sensing data. This limits their generalizability in complex scenarios. To address these challenges, we propose a novel remote sensing counting network that effectively captures informative signals from relatively limited annotated data. Specifically, we first introduce a graph-driven input regulator (GdIR) that constructs a graph structure by modeling relationships among input features, effectively capturing intrinsic contextual dependencies. This structure allows the regulator to assign adaptive pixel-level weights to network inputs, prioritizing relevant signals while mitigating the risk of overfitting to a fixed data distribution. Second, we design a dynamic region-aware module (DRaM) that leverages fuzzy logic to adaptively identify and enhance highly discriminative local regions. In this way, it improves the robustness of the feature representations. Extensive experiments demonstrate the effectiveness of the proposed method compared with several state-of-the-art methods.**

***Index Terms*—Fuzzy rule, object counting, regulator, remote sensing.**

I. Introduction

REMOTE sensing counting aims to automatically estimate the number of a specific type of objects (e.g., vehicles, buildings, and ships) in high-resolution aerial or satellite images, providing crucial support for decision-making, resource allocation, and monitoring of dynamic changes. This technology holds significant application value in various fields, including urban planning, traffic surveillance, agricultural assessment, environmental protection, and disaster response. However, achieving accurate prediction remains challenging due to factors such as large variations in object scale, dense distribution of objects, and cluttered backgrounds.

Many recent studies have leveraged the powerful modeling capabilities of pre-trained models, particularly those based on deep convolutional networks [1], [2] and vision Transformers [3], [4]. These models, typically pre-trained on large-scale datasets such as ImageNet [5], learn rich and highly generalizable feature representations that effectively capture fundamental visual concepts. When fine-tuned on downstream tasks like remote sensing counting, they often achieve significantly better performance than models trained from scratch. This transfer of knowledge enhances the model's ability to recognize remote sensing targets in complex scenarios.

However, most existing methods are empirically designed and tend to overlook a fundamental challenge: the limited training data. The annotation of remote sensing counting images is both time-consuming and costly, leading to a scarcity of high-quality annotated datasets. This limitation severely hampers the model's ability to generalize to real-world scenarios involving variations in lighting, object scale, and complex background interference. While pre-training can partially mitigate this issue, the effectiveness of fine-tuning remains constrained by the size of the available labeled data. Thus, when the model encounters distribution shifts during testing, its performance degrades significantly. At the root of this issue, the limited data fails to reflect the underlying distribution patterns of complex scenes, making the network easily susceptible to noise and irrelevant details during training. This often leads to overfitting and prevents the model from effectively learning the essential feature representations of the target, ultimately limiting its performance.

To address the aforementioned challenge, we propose an input-regulated remote sensing counting network. Specifically, we propose a graph-driven input regulator (GdIR) that constructs a graph structure from the features of each input sample. This structure serves as conditional information to adaptively modulate the pixel values of inputs by guiding learnable weight parameters, referred to as regulators. By integrating the graph structure with these dynamic parameters,

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the network achieves effective self-regulation based on the input content, thereby reducing the risk of overfitting to limited training samples. In addition, we design a dynamic region-aware module (DRaM) to further strengthen the model's focus on informative local regions. It utilizes fuzzy rules to analyze the relationships between local regions in feature maps, identifying and enhancing discriminative regions to improve the model's representation capability. Extensive experiments are conducted to validate the effectiveness of the proposed method. Results demonstrate that our method consistently outperforms existing methods and achieves state-of-the-art performance across multiple remote sensing datasets. The main contributions are summarized as follows.

1) We propose a novel network framework to enhance the model's ability to capture discriminative features under limited training data.
2) We introduce a GdIR, which utilizes the topological relationships within input embeddings to steer the network's attention toward critical input signals.
3) We propose a DRaM that enhances the network's representation capability by identifying key regions using fuzzy rules.
4) The proposed method is compatible with both mainstream density map-based and point-based frameworks. Extensive experiments across both types of frameworks demonstrate its effectiveness, consistently achieving state-of-the-art performance.

II. Related Work

A. Surveillance Perspective Object Counting

Early research on object counting primarily targeted surveillance scenarios, such as crowd counting at transportation hubs and vehicle counting in congested areas. These tasks often involve complex images with severe occlusions, varying object scales, and dense object distributions. With the advent of deep learning, deep neural networks have emerged as the dominant approach for object counting, owing to their exceptional image modeling capabilities. For example, CSRNet [6] adopted a pretrained VGG-16 network as the front-end to extract spatial features, followed by a dilated convolutional layer as the back-end to capture more contextual information. Similarly, Liu et al. [7] presented an end-to-end trainable deep learning framework that adaptively captures contextual information by leveraging multiple convolutional kernels with varying receptive field sizes.

Besides handling the scale variations of the object, the target in the surveillance view also exhibits serious occlusion and background interference problems. In this regard, Chen et al. [8] proposed a masked feature prediction module and a supervised pixel-level contrastive learning module to enhance the model's capability in distinguishing between foreground and background. Zhai et al. [9] presented a unified framework for simultaneous object counting and localization. They designed a scale-aware module with local and global dual branches to capture multiscale information, along with a context-aware module that leverages channel and spatial attention to suppress background interference. Recently, Shi et al. [10] explored a foreground distillation method that leveraged point annotations to generate foreground masks and trained an auxiliary network to learn foreground representations, aiming to mitigate background interference.

Since CNNs struggle to model global dependencies between features, the self-attention mechanism in Transformers has been introduced to effectively capture such dependencies, thereby addressing the limitations of CNNs. As a result, numerous Transformer-based object counting models have been proposed. Jiang et al. [11] introduced a density attention network and an attention scaling network to alleviate performance discrepancies across different regions of the model. Lin et al. [12] proposed to adjust the attention and input node features based on two different types of graphs to alleviate the homogeneity problem of the attention maps of the patches in the Transformer. In addition, to alleviate the annotation burden for dense object scenes, Xiong et al. [13] proposed converting the regression problem into a ranking prediction task using a weakly supervised approach that leverages binary rankings between highly contrasting image pairs as training signals. Lin et al. [14] introduced an agency-guided semi-supervised counting model that uses image density information to guide learning, achieving strong performance among semi-supervised counting methods.

B. Remote Sensing Object Counting

Compared to the relatively fixed perspectives of surveillance imagery, remote sensing offers greater flexibility and a wider range of application scenarios, enabled by rapid advances in satellite and UAV technologies. Consequently, research in remote sensing object counting has seen significant growth in recent years. Gao et al. [15] introduced a benchmark for remote sensing object counting that integrates channel and spatial attention mechanisms to emphasize salient image regions. To address the challenge of scale variation, Gao et al. [16] proposed a method that adaptively extracts multiscale features and integrates global context to guide the model in selecting the most suitable scale. Subsequently, TASNet [17] was developed to mitigate the effects of cluttered backgrounds by introducing a triple attention mechanism that jointly considers spatial, channel, and scale attention.

To further enhance discriminative ability, GGANet [18] combined group channel attention with learnable graph attention, reducing the influence of irrelevant pixels. Recently, BDRNet [19] introduced an adaptive kernel-weighted attention mechanism to learn the optimal weights for dilated convolutions with varying receptive field sizes, thereby improving the semantic understanding capabilities of the network. Gao et al. [20] and Cui et al. [21] proposed a network specialized for multi-category object counting in remote sensing scenarios to facilitate research in this area. Building on this foundation, Mamba-MOC [22] developed a multi-category object counting network based on the Mamba architecture. To address the inherent limitations of causal scanning in the Mamba model, the network incorporates a contextual state space module that effectively captures both global and local contextual information. OnKLnet [23] integrated a two-stage distillation strategy into a unified single-stage framework and

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further enhanced knowledge transfer efficiency through a new relational distillation.

These methods typically leverage pretrained models to improve performance on limited remote sensing data [24]. However, the inherent scarcity of remote sensing data remains a fundamental constraint often overlooked in efforts to enhance model performance. Few-shot counting methods offer a straightforward solution, but their task settings typically rely on an extremely limited number of training samples (e.g., one-shot or five-shot) [25]. Despite efforts to enhance performance through strategies such as domain adaptation [26] and pseudo-label learning [27], [28], the severe scarcity of training data remains a fundamental bottleneck that constrains further improvement. Therefore, effectively extracting and utilizing key information from remote sensing data to improve model generalization represents a crucial and valuable research direction for further investigation.

C. Fuzzy Rules

Fuzzy systems [29] introduce the concepts of fuzzy sets and fuzzy inference, enabling effective handling of data uncertainty and ambiguity through the learning of fuzzy rules. Such systems represent the fuzziness of input data with fuzzy sets and perform inference based on defined IF-THEN fuzzy rules. When combined with neural networks, they form neuro-fuzzy models that leverage the strengths of both approaches. In these models, fuzzy systems interpret ambiguous and uncertain features extracted by neural networks, thereby enhancing the network's ability to produce robust feature representations. Consequently, neuro-fuzzy models have been widely applied to various visual tasks, including classification and regression. For example, Wang et al. [30] integrated fuzzy systems with deep neural networks to address sentiment analysis, achieving complementary representation learning by extracting and fusing multimodal features, while leveraging fuzzy logic to better manage the inherent ambiguity in sentiment expression. Huang et al. [31] proposed a novel separable convolutional neural network incorporating fuzzy channel filtering to enhance performance in medical image recognition. Sajid et al. [32] combined a random vector functional link network with a neuro-fuzzy system, employing three distinct clustering techniques to determine the centers of the fuzzy layer, thereby improving the model's generalization capability in both regression and classification tasks. In addition, fuzzy systems are based on a set of IF-THEN fuzzy rules, which can be formally defined as

$$\begin{aligned} &\text{IF } (x_1 \text{ is } u_{k1}) \text{ AND } \ldots \text{ AND } (x_m \text{ is } u_{\mathrm{km}}) \\ &\text{THEN } y_k = \mu_k(x_1, x_2, \ldots, x_m), \quad k = 1, 2, \ldots, K \end{aligned} \tag{1}$$

where K denotes the number of rules, $u_{ki}(i = 1, \ldots, m)$ is a fuzzy set defined according to the input feature vector $x = [x_1, \ldots, x_m]$, m denotes the number of inputs. The function μ_k denotes any correlation function that can correctly characterize the output within the given fuzzy rules.

In this study, we introduce fuzzy rules into remote sensing counting for the first time to enhance the model's capacity to perceive key local information within images. To address common challenges such as uneven object distribution and complex background interference, the integration of fuzzy rules not only strengthens the model's focus on discriminative regions but also enhances the interpretability of its decision-making process. Specifically, we employ a Gaussian membership function to compute the membership degree of each local region based on different fuzzy rules. This membership degree serves as an indicator of regional importance, guiding the model to identify more discriminative feature regions, thereby improving its overall representational capability.

III. Proposed Method

In this work, we propose an input-regulated remote sensing counting network. The overall framework is depicted in Fig. 1, consisting of four main components: a backbone, a GdIR, a DRaM, and a regression head. Specifically, the GdIR is first used to capture intrinsic relationships within the input data, guiding the network to more effectively capture the critical distribution of inputs. Next, a Mamba-based backbone is employed to extract hierarchical features, which are then aggregated to form multi-scale representations. The DRaM is subsequently applied to learn region-aware features, where fuzzy rules assign adaptive weights to different regions, refining the feature representation. Finally, the regression head estimates the counting results.

A. Backbone

We adopt VMamba [33] as the feature extractor, a vision backbone characterized by linear computational complexity. Compared to traditional Transformer-based networks, VMamba significantly reduces computational cost while retaining the ability to capture global dependencies. It employs a multistage architecture, where each stage progressively increases the number of channels while reducing the spatial resolution of feature maps. As the network deepens, its capacity to represent input images abstractly improves, but often at the expense of fine-grained details.

To preserve information of different granularities, we employ multilevel feature aggregation to enhance the model's ability to perceive multiscale objects. Specifically, features are extracted from three distinct stages and upsampled to 1/8 of the original input resolution, denoted as $X_j \in \mathbb{R}^{C_j \times H/8 \times W/8}$, where $j \in 1, 2, 3$. These features are concatenated along the channel dimension to form $\hat{X} = [X_1, X_2, X_3]$, where $[\cdot]$ denotes the concatenation operation. While this aggregated representation offers enhanced expressive capacity, it may also introduce redundancy, making it more challenging to identify effective features and increasing computational overhead. To mitigate this, we apply convolutional operations to compress the channel dimensions, thereby enhancing the efficiency of subsequent processing. This operation can be formally expressed as

$$\mathcal{K} = \mathrm{ReLU}\left(\mathrm{BN}\left(\psi_1\left(\hat{X}\right)\right)\right) \tag{2}$$

where $\psi_1(\cdot)$ denotes convolutional operations, $\mathrm{BN}(\cdot)$ denotes BatchNorm, and $\mathrm{ReLU}(\cdot)$ denotes the ReLU activation function.

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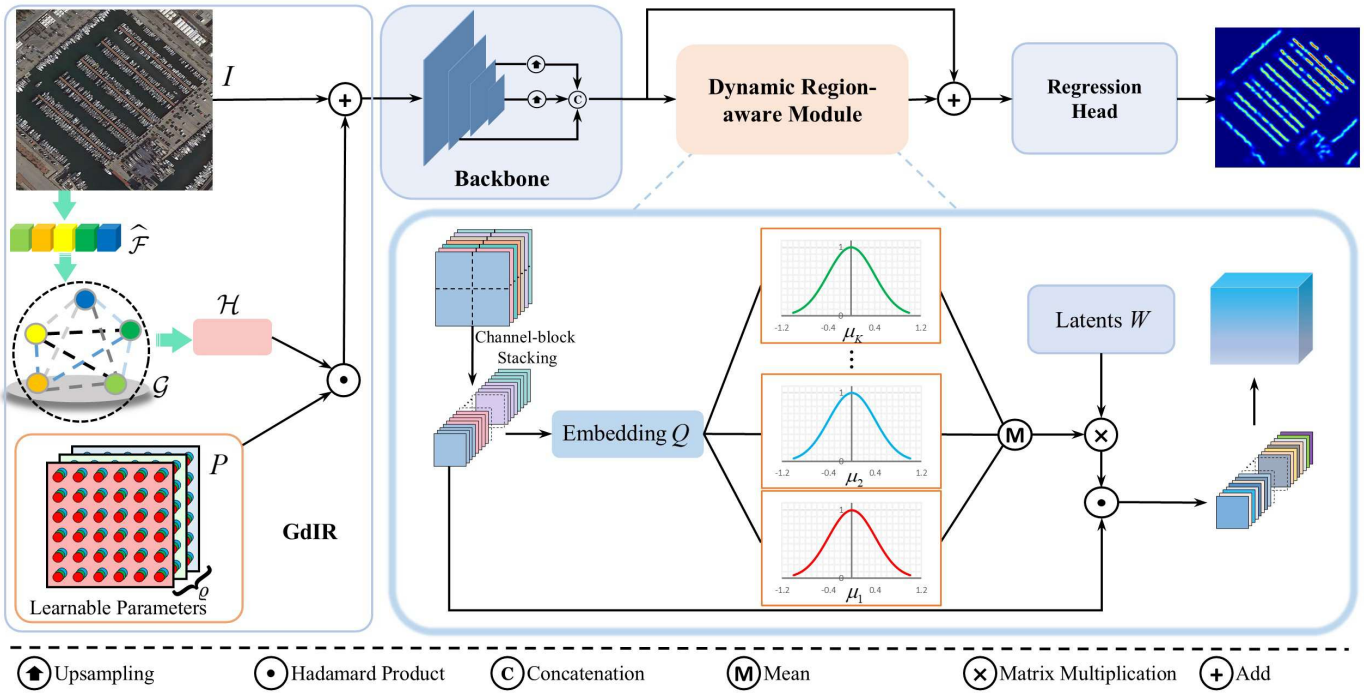


Fig. 1. Overall framework of our proposed network. The network consists of two primary components: a GdIR (Section III-B) and a DRaM (Section III-C). The former is designed to adjust the input distribution, while the latter enhances the network's representation capability by deliberately integrating fuzzy rules. $P = \{P_1, P_2, \ldots, P_\varrho\}$ denotes the learnable regulator with the same dimension as the network input I, which consists of learnable parameters.

B. Graph-Driven Input Regulator

In computer vision, many tasks such as image classification and video understanding typically benefit from large amounts of annotated training data. In contrast, remote sensing counting tasks often suffer from severe limitations in annotated training samples. This scarcity makes the model more susceptible to overfitting, amplifying the influence of noise and minor details, and ultimately constraining its generalization ability. To this end, we propose a GdIR to enhance the learning of critical input information. The core idea of the module is to leverage the rich structured information inherent in each input to guide the regulator in adjusting the input distribution. Our method functions as an adaptive, graph-structured pre-filter that prioritizes task-relevant information. By intervening at the input stage, it steers the learning dynamics of the entire network, reducing dependence on a fixed training distribution and ultimately improving the model's generalization capability.

It first dynamically constructs a graph based on the relationships among feature embeddings of the image, effectively capturing internal contextual associations and topological structures. Specifically, an adaptive average pooling operation is first applied as a low-pass filter to downsample the original input image, effectively suppressing high-frequency noise while improving computational efficiency. Subsequently, convolutional operations are employed to extract the fundamental features of the image, followed by normalization using a sigmoid function. This computational process can be formulated as

$$\begin{cases} \mathcal{B} = \text{AdaPool}\,(I) \\ \mathcal{F} = \text{Sigmoid}\,(\psi_2\,(\mathcal{B})) \end{cases} \tag{3}$$

where AdaPool$(\cdot)$ denotes the adaptive average pooling operation, $\psi_2(\cdot)$ indicates convolutional operations, and I means the network input.

Given the initial features $\mathcal{F} \in \mathbb{R}^{\vec{C}\times\vec{H}\times\vec{W}}$, we represent it with an undirected graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$ in which $\mathcal{N}$ and $\mathcal{E}$ denote a finite set of nodes and a finite set of edges, respectively. For each node $n_i \in \mathcal{N}$, its representation $r_i = \hat{\mathcal{F}}[i]$, where $\hat{\mathcal{F}} \in \mathbb{R}^{L\times\vec{C}}$ is derived by flattening $\mathcal{F}$ along the spatial dimension. The edges e_{ij} are modeled by $e_{ij} = |r_i - r_j|$, and then we have the edge embedding $E = (e_{ij})_{L\times L\times\vec{C}}$. It is evident that the edges in the graph allow connections between any pair of nodes, enabling the model to capture long-range dependencies.

Then, the graph structure is leveraged to adaptively select a set of learnable regulators highly relevant to the network input. These regulators serve as adaptive supplements to the network inputs, enriching the diversity of the input distribution and guiding the network to focus on more salient relational patterns. Specifically, taking the average of the rows of the edge embedding yields $g \in \mathbb{R}^{L\times\vec{C}}$. Then, we independently process this output through two separate linear layers. The resulting features are aggregated along the channel dimension, followed by a linear layer to adjust the dimensionality of the fused features. Finally, the adaptive weights are computed by applying the softmax function, which enables a weighted linear combination of the predefined regulators. The entire process can be formulated as

$$\begin{cases} g = \text{Mean}\,(E) \\ \mathcal{H} = \text{Softmax}\left(\left[\vartheta\,(\kappa_1\,(g))\,\vartheta\,(\kappa_2\,(g))\right]\right) \\ \mathcal{R} = I + \sum_{i=1}^{\varrho} \mathcal{H}_i \odot P_i \end{cases} \tag{4}$$

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where $\kappa_1(\cdot)$ and $\kappa_2(\cdot)$ denote the linear operations, respectively. ϑ indicates the dimension transformation, $\odot$ denotes the elementwise product. P_i denotes the ith learnable regulator with the same dimension as the input I. In doing so, the network reduces its reliance on shallow or purely local features under limited data conditions, avoids being trapped in suboptimal local minima during optimization.

C. Dynamic Region-Aware Module

To address the challenges of uneven target distribution and complex background interference in remote sensing images, we further propose a DRaM. It generates dynamic coefficients for each partitioned region in the feature map based on fuzzy rules, followed by a weighted fusion process. This design is motivated by the observation that the visual discriminability of local regions is often not binary, i.e., "important" versus "unimportant," but instead exhibits substantial ambiguity. For instance, object boundaries may become unclear due to occlusion or lighting variations, and certain features may degrade partially. To handle such uncertainties, fuzzy logic leverages continuous membership functions and smooth inference processes, allowing it to adjust regional features adaptively while avoiding abrupt weight shifts. This could lead to more stable and reliable feature representations.

Specifically, we first divide the feature maps $\mathcal{Z}$ from the high-level semantic layer into N nonoverlapping regions (or patches), denoted as $\mathcal{Z} = \{z_1, z_2, \ldots, z_N\}$. We then propose using channel-block stacking to aggregate the features of all regions, namely, the blocks from each channel are first concatenated in sequence, followed by the sequential aggregation of all channels. In this approach, the blocks from each channel are first concatenated in sequence, and then the channels are aggregated sequentially. This method preserves channel continuity and the relative spatial order, thereby facilitating more effective integration of global information. Subsequently, a compact global embedding Q is generated through adaptive average pooling. Taking this embedding as input, the following fuzzy operations are performed. First, the Gaussian membership function is used to compute the degree of membership of Q to each fuzzy rule, namely

$$\mu_{ki} = \exp\left[-\frac{(q_i - c_{ki})^2}{2\sigma^2}\right] \tag{5}$$

where c_{ki} and σ represent the center and width of the Gaussian function, respectively, c_{ki} is a learnable parameter, σ is a hyper-parameter, and q_i denotes the ith element of Q. Next, a normalization operation is applied to balance the contribution weights of different rules

$$\mu_i = \frac{\sum_{k=1}^{K} \mu_{ki}}{K} \tag{6}$$

where μ_i denotes the membership degree of the ith region of $\mathcal{Z}$. Furthermore, latents are introduced to learn membership embeddings, which facilitate the discovery of intrinsic relationships among normalized fuzzy rules. These embeddings are subsequently used to weight the regions, as formulated below

$$\begin{cases} \mu_i' = \mu \otimes W_i \\ h_i = \mu_i' z_i + z_i \end{cases} \tag{7}$$

where $\mu = (\mu_i)_{1\times\hbar}$, W_i indicates the learnable latents, $\otimes$ denotes the matrix multiplication. The weighted regional features are reshaped to match the original dimensions of $\mathcal{Z}$ before being fed into the regression head.

IV. Experiments

In this section, we first present the experimental setup, including implementation details, datasets, and evaluation metrics. We then compare our method with several recent state-of-the-art methods, demonstrating its superior performance. Finally, we conduct ablation studies to validate the effectiveness of the proposed components.

A. Experiment Setup

1) Implementation Details: All experiments are conducted using the PyTorch framework on a GeForce RTX 4090 GPU. We adopt VMamba-S [33] as the backbone, pre-trained on ImageNet [34]. As the proposed method functions as a feature extractor, it can be integrated into various counting frameworks. To demonstrate its versatility, we apply it to two representative frameworks: one based on point regression [35] and the other on density maps [36]. This enables a comprehensive evaluation of the proposed network across diverse application scenarios.

To accommodate the distinct requirements of each framework, we adopt experimental settings aligned with prior work. For the point-based framework, input images are randomly scaled to 0.7–1.3 times their original size and then randomly cropped into four 128×128 patches. Data augmentation includes horizontal flipping with a 50% probability and image normalization. The model is trained using the Adam optimizer with an initial learning rate of 1e-4 and a weight decay of 1e-4. As for the density map-based framework, we apply random cropping with a fixed size and use horizontal flipping with a 50% probability for augmentation. The network is trained using the AdamW optimizer, also with an initial learning rate of 1e-4 and a weight decay of 1e-4.

2) Datasets and Metrics: We evaluate our method using three datasets: RSOC [15], CARPK [37], and PUCPR+ [37]. Below is a detailed description of each dataset and their corresponding training, validation, and testing splits. The RSOC dataset contains four categories of satellite images: building, small-vehicle, large-vehicle, and ship. It consists of 2468 building images, 280 small vehicle images, 172 large vehicle images, and 137 ship images. Specifically, building has 964 training images, 241 validation images, and 1263 test images. Small-vehicle contains 200 training images, 22 validation images, and 58 test images. Large-vehicle includes 98 training images, ten validation images, and 64 test images. The ship has 88 training images, nine validation images, and 40 test images.

Furthermore, the CARPK dataset consists of 1448 aerial images containing approximately 89 777 cars. It is split into 989 training images and 459 test images. The PUCPR+ dataset contains 125 images with a total of about 17 000 cars. It is divided into 100 training images and 25 test images.

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TABLE I
COMPARISON OF OUR METHOD WITH OTHER METHODS ON RSOC. THE BEST PERFORMANCE IS IN BOLDFACE, AND THE SECOND BEST IS UNDERLINED

Method	Params(M)	FLOPs(G)	Building		Small-vehicle		Large-vehicle		Ship		Average	
			MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
BL [38]	21.5	26.99	28.95	32.93	185.79	703.31	26.03	51.54	64.15	86.01	76.23	218.45
ASPD [15]	22.7	37.96	7.59	10.66	433.23	1238.61	42.87	63.49	216.27	352.23	174.99	416.25
DM-Count [36]	21.5	26.99	7.22	10.6	189.6	736.63	17.05	24.94	65.73	92.63	69.9	216.2
P2PNet [35]	19.2	23.29	7.46	10.34	174	666.44	**11.61**	16.94	62	76.43	63.77	192.54
DRSOC [39]	-	-	7.93	11.82	238.46	625.9	12.94	20.25	50.45	**65.24**	77.45	180.8
PET [40]	20.91	28.42	10.78	13.96	419.74	1212.18	36.39	54.52	258.83	396.39	181.44	419.26
LMSFFNet [4]	9.49	-	6.33	9.38	205.03	**503.47**	13.15	29.47	57.15	80	70.42	**155.58**
PML [41]	16.26	21.65	8.3	12.78	217.17	652.98	32.15	48.2	139.58	172.17	99.3	221.53
GCFL [42]	21.51	27.01	7.79	10.74	222.4	768.83	16.3	23.31	55.32	77.05	75.4	219.92
FFNet [43]	34.58	6.12	6.02	**8.66**	215.83	858.21	23.9	36.37	68.51	102.71	78.57	251.49
OnKLNet [23]	0.76	4.00	7.58	10.62	169.86	613.3	12.27	**16.7**	81.8	105.57	67.88	186.55
Ours + DM Loss	48.95	12.13	**5.92**	8.81	194.43	745.88	12.56	22.27	51.98	72.21	66.22	212.29
Ours + P2P Loss	49.73	12.93	8.04	12.06	**147.82**	571.16	12.14	19.5	**49.4**	67.95	**54.35**	167.67

The results are evaluated using two metrics: mean absolute error (MAE) and mean squared error (mse)

$$\text{MAE} = \frac{1}{\Gamma}\sum_{i=1}^{\Gamma}|y_i - \hat{y}_i| \tag{8}$$

and

$$\text{mse} = \sqrt{\frac{1}{\Gamma}\sum_{i=1}^{\Gamma}(y_i - \hat{y}_i)^2} \tag{9}$$

where Γ is the number of test images, y_i and $\hat{y}_i$ are the predicted values and ground truth of the ith image, respectively. We also report the number of parameters (Params) and floating-point operations per second (FLOPs) to characterize the model's computing complexity, using an input resolution of 256×256.

B. Comparison With Other Methods

Table I presents the comparison results with some SOTA methods on the RSOC dataset, including BL [38], DM-Count [36], P2PNet [35], DRSOC [39], ASPD [15], PET [40], LMSFFNet [4], PML [41], GCFL [42], OnKLNet [23], and FFNet [43]. As shown, our method achieves state-of-the-art performance across two mainstream frameworks, with particularly strong results in the point regression framework, where it attains superior average counting accuracy across different categories. Compared to the baseline DM-Count [36], our approach yields significant improvements in most categories, except for small-vehicle, where the MAE slightly increases by 0.3%. Against P2PNet [35], our method achieves substantial MAE reductions of 15.0% and 20.3% on the small-vehicle and ship datasets, respectively, though it shows a slight performance drop on building and large-vehicle. We attribute this to the high visual similarity between large-sized objects and the background, which may hinder the effectiveness of the distribution augmentation strategy in capturing discriminative features, thereby increasing the learning difficulty. Notably, compared to recent methods such as FFNet [43] and OnKLNet [23], our method maintains consistent advantages across multiple dataset subsets, demonstrating its robustness in remote sensing object counting tasks. Although our method achieves highly competitive performance, its parameters are higher than those of some existing SOTA methods. Note that our model maintains relatively low FLOPs, demonstrating high computational efficiency during inference.

TABLE II
COMPARISON OF OUR METHOD WITH OTHER METHODS ON CARPK AND PUCPR+

Method	CARPK		PUCPR+	
	MAE	MSE	MAE	MSE
MCNN [44]	39.1	43.3	21.86	29.53
CSRNet [6]	11.48	13.32	8.65	10.24
RetinaNet [45]	16.63	22.3	24.58	33.46
BL [38]	9.58	11.38	6.54	8.13
PSGCNet [16]	8.15	10.46	5.24	7.36
LMSFFNet [4]	7.05	9.03	4.49	6.21
GGANet [18]	10.07	12.9	1.71	2.3
BDRNet [19]	5.94	8.59	1.91	2.49
EdgeCount-T [46]	6.11	8.51	3.56	4.72
Ours + DM Loss	**5.52**	8.01	**1.44**	**1.93**
Ours + P2P Loss	5.92	**7.95**	1.92	2.51

Furthermore, Table II presents the comparison results with other methods on the CARPK and PUCPR+ datasets. As shown, our methods achieve SOTA performance across all evaluation metrics. Among the variants, the model trained with DM Loss delivers the best overall results. While the version using P2P Loss performs slightly worse, the difference is marginal. Notably, it offers the additional benefit of providing explicit location information, which is valuable for downstream target analysis. These quantitative results demonstrate that our method achieves highly competitive performance

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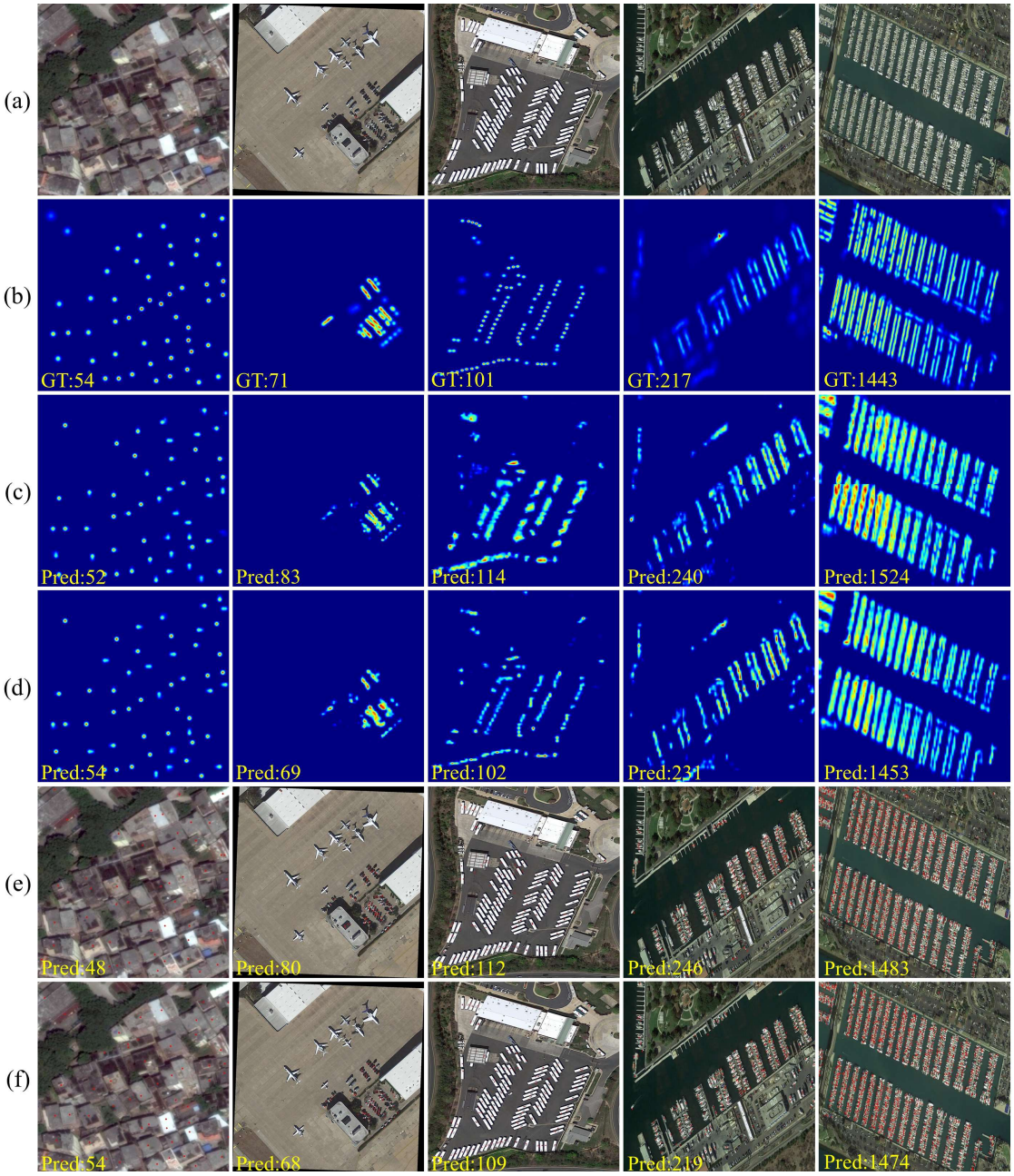


Fig. 2. Visual comparison of our method with other methods. (a) and (b) Input image and the corresponding ground truth, respectively. (c) and (d) Results of FFNet [43] and our method with DM Loss, both of which are based on the density map framework. (e) and (f) Outputs of OnKLNet [23] and our method with P2P Loss, respectively, which follow the point-based framework.

across different network frameworks and adapts effectively to diverse remote sensing scenarios, thereby validating both its effectiveness and generalization capability.

To more intuitively validate the effectiveness of our method, we visualize the comparison results with other methods under different density scenarios, as shown in Fig. 2. The results based on the density map framework and the point counting framework are displayed separately. The visualization clearly indicates that density map-based methods can effectively present the overall distribution of targets and clearly reflect their sparsity or density, which is significant for resource allocation. This method can monitor the aggregation of targets in real-time, promptly identify potential congestion issues, and thereby prevent the negative impacts caused by excessive target aggregation. In contrast, the point-based method can regress the precise location information of targets, enabling accurate localization of individual instances, which facilitates finer-grained planning and subsequent location-based processing and analysis. Furthermore, the comparison results show that our method significantly outperforms recent counting methods, FFNet [43] and OnKLNet [23], under both frameworks. Although there are slight fluctuations in the counting results across the two frameworks, all predictions are very close to the ground truth. In summary, these visual results

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TABLE III

COMPARISON WITH OTHER METHODS ON LESS TRAINING DATA. THE BEST PERFORMANCE IS IN BOLDFACE, AND THE SECOND BEST IS UNDERLINED

Method	Percentage	Average	
		MAE	MSE
DM-Count [36]	25%	<u>94.53</u>	<u>321.68</u>
FFNet [43]		137.7	373.68
Ours + DM Loss		97.88	326.1
Ours + P2P Loss		**90**	**268.42**
DM-Count [36]	50%	<u>87.56</u>	291.37
FFNet [43]		97.1	265.33
Ours + DM Loss		94.3	<u>247.34</u>
Ours + P2P Loss		**65.91**	**164.21**
DM-Count [36]	100%	69.9	216.2
FFNet [43]		78.57	251.49
Ours + DM Loss		<u>66.22</u>	<u>212.29</u>
Ours + P2P Loss		**54.35**	**167.67**

TABLE IV

ABLATION STUDY ON KEY COMPONENTS OF OUR NETWORK. GDIR AND DRAM DENOTE THE GDIR AND DRAM, RESPECTIVELY

Method	Large-vehicle		SHA	
	MAE	MSE	MAE	MSE
Baseline	16.46	27.98	54.29	97.16
Baseline + GdIR	13.68	21.46	51.01	81.17
Baseline + GdIR + DRaM	12.56	22.27	49.53	77.05

TABLE V

ABLATION STUDY ON THE NUMBER OF LEARNABLE REGULATORS

ϱ	Large-vehicle		SHA	
	MAE	MSE	MAE	MSE
1	16.9	25.05	55.21	91.77
3	16.67	26.88	53.85	96.47
5	**13.68**	**21.46**	**51.01**	**81.17**
7	14.81	26.32	52.71	87.77
9	15.42	24.37	54.56	90.31

TABLE VI

ABLATION STUDY ON THE REPRESENTATION OF GRAPH EDGES

Method	Large-vehicle		SHA	
	MAE	MSE	MAE	MSE
Absolute Error	**13.68**	**21.46**	**51.01**	**81.17**
Squared Error	14.36	26.88	52.43	82.65
Root Squared Error	15.78	25.51	51.97	87.52

further strongly confirm the effectiveness of the proposed method from a perceptual perspective.

To further validate the effectiveness of the proposed method, we evaluate model performance under different proportions of annotated training data. Table III shows that performance generally declines as the amount of annotated data decreases. However, when the training data are reduced by half, the P2P loss-based version still achieves average performance on RSOC that surpasses several comparative methods trained with the full dataset. This demonstrates that our method can effectively extract intrinsic knowledge even with limited annotations. Furthermore, when the training data are reduced to one quarter, the model attains an MAE of 90.00, lower than that of other methods, while maintaining substantially better mse performance. These results further confirm the robustness of our method in data-limited scenarios.

C. Ablation Studies

We conduct ablation studies on large-vehicle and ShanghaiTech Part_A (SHA) [44] to evaluate the effectiveness of our proposed method.

1) Ablation Studies on the Key Proposed Components: Our proposed method integrates two key modules: the GdIR and DRaM. To evaluate their effectiveness, we conduct detailed ablation studies, as shown in Table IV. Before discussing the results, we briefly describe the baseline, which uses a multiscale aggregation backbone for feature extraction and a regression head for density estimation. The results show that introducing the GdIR module significantly enhances performance on large-vehicle, decreasing MAE by 16.9% and mse by 23.3%, and on SHA, reducing MAE by 6% and mse by 16.5% compared to the baseline. This highlights how the adjustment strategy of input distribution contributes to improving the network's capability in uncovering the essential and meaningful representations inherent in the data. Adding DRaM leads to further gains on large-vehicle, reducing the MAE to 12.56 while maintaining a comparable mse. Under the same setting on SHA, both MAE and mse exhibit a notable reduction. These improvements indicate that DRaM strengthens the model's ability to capture regional features, thereby enhancing discriminative performance in complex scenarios. Overall, the ablation results clearly demonstrate the effectiveness of both GdIR and DRaM.

2) Ablation Studies on GdIR: We conduct an ablation study on the number of learnable regulators in the GdIR to evaluate its effect on network performance. As shown in Table V, the network achieves optimal results when the number of regulators is set to 5. Both increasing and decreasing this number lead to performance degradation. This behavior can be attributed to the model's representational capacity and optimization complexity. When too few regulators are used, the network struggles to generate sufficiently diverse distribution representations. Conversely, an excessive number of regulators increases the difficulty of identifying the optimal distribution and introduces additional optimization overhead. Based on this analysis, we fix the number of learnable regulators to 5 in all subsequent experiments.

In graph structure modeling, edge construction depends heavily on how relationships between nodes are measured, and the choice of measurement method has a direct impact on network performance. To determine the most effective strategy, we compare three commonly used approaches for relationship modeling, as shown in Table VI. Experimental results indicate

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TABLE VII

COMPARISON WITH RECENT GRAPH-BASED WEIGHTING METHODS. THE BEST PERFORMANCE IS IN BOLDFACE, AND THE SECOND BEST IS UNDERLINED

Method	Large-vehicle		SHA	
	MAE	MSE	MAE	MSE
AGCA [47]	16.16	24.13	<u>53.8</u>	<u>92.36</u>
BIQA [48]	<u>13.56</u>	<u>22.54</u>	55.61	97.68
Ours	**12.56**	**22.27**	**49.53**	**77.05**

TABLE VIII

ABLATION STUDY ON THE NUMBER OF REGIONS

Number	Large-vehicle		SHA	
	MAE	MSE	MAE	MSE
1	13.31	23.64	50.04	82.65
4	**12.56**	**22.27**	**49.53**	**77.05**
16	12.69	22.17	53.27	90.01
32	14.44	22.58	51.97	90.01

TABLE IX

ABLATION STUDY ON STACKING METHODS FOR FEATURE BLOCK PARTITIONING

Method	Large-vehicle		SHA	
	MAE	MSE	MAE	MSE
Block-channel stacking	14.15	23.32	50.75	82.36
Channel-block stacking	**12.56**	**22.27**	**49.53**	**77.05**

TABLE X

ABLATION STUDY ON THE IMPACT OF THE NUMBER OF FUZZY RULES

K	Large-vehicle		SHA	
	MAE	MSE	MAE	MSE
1	14.53	22.89	50.9	84.8
2	**12.56**	**22.27**	**49.53**	**77.05**
3	15.29	25.89	51.89	81.48
4	14.67	25.39	50.61	85.53
5	14.34	25.66	53.07	85.95

that using absolute error to measure node relationships yields significantly better performance than both squared error and root mean squared error. We attribute this to the robustness of absolute error against outliers, which allows it to more reliably capture node associations in the presence of noise. This also explains why root squared error outperforms squared error, as it mitigates the amplification of large deviations caused by squaring. Based on these results, we adopt the absolute error as our final solution.

To further highlight the advantages of our method, we compare it with two recent graph-based weighting methods, AGCA [47] and BIQA [48], as shown in Table VII. The results show that our method achieves a clear performance advantage. Specifically, AGCA constructs feature graphs along the channel dimension, which leads to the loss of crucial spatial information. BIQA, on the other hand, builds adjacency matrices based on connections between adjacent nodes and self-relations, but fails to capture deeper dependencies among internal feature nodes. In contrast, our method leverages the spatial information of input features to establish node relationships, enabling the regularization mechanism to adapt to the actual data distribution. This allows the network to focus more effectively on the most informative signals of the input.

3) Ablation Studies on DRaM: Building upon Baseline 2 (i.e., Baseline + GdIR), we conduct ablation studies to evaluate key configurations of the DRaM. This module operates by partitioning the feature maps produced by the feature extractor into spatial regions and performing region-specific processing. We first investigate the effect of the number of partitions on model performance, as shown in Table VIII. The results show that the model achieves optimal performance on both MAE and mse metrics when the number of regions is set to 4. Increasing the number of regions beyond this point leads to significant performance degradation relative to Baseline 2. We attribute this to the excessive granularity of region partitioning, which increases the complexity of regional channel representations and makes it more difficult for the network to identify salient regions effectively. Interestingly, when the number of regions is set to 1 (i.e., no partitioning), the model's MAE slightly improves over Baseline 2, while the mse shows a minor increase. This suggests that the fuzzy rule-based weighting strategy does not adversely affect the original feature extraction process. In contrast, setting the number of regions to four results in clear improvements in both metrics, thereby confirming the effectiveness of the proposed strategy.

We further investigate how different stacking strategies for spatially partitioned feature maps affect network performance. As illustrated in Fig. 3, we compare two approaches as follows.

1) *Block-Channel Stacking:* The feature maps are first divided into spatial blocks, where each block retains all channel information for its corresponding location. These blocks are then stacked along the channel dimension. This strategy preserves channel interactions within local regions but diminishes channel associations across different spatial areas.
2) *Channel-Block Stacking:* Feature maps are divided channelwise, and each individual channel is further partitioned into spatial blocks. All resulting blocks across channels are then stacked along the channel dimension. This method emphasizes channel continuity and preserves the spatial order within each channel, thereby promoting more efficient integration of global contextual information. As shown in Table IX, the channel-block stacking method achieves superior performance. We attribute this to its ability to leverage global information when constructing fuzzy rules, allowing the network to generate more effective weighting strategies and better exploit spatial feature representations.

We analyze the effect of the number of fuzzy rules on the fuzzy reasoning process. In a fuzzy system, each input feature calculates its degree of membership across different rules, and the contributions of these rules are then aggregated. Consequently, the number of fuzzy rules plays a critical role

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TABLE XI

COMPARISON WITH RECENT WEIGHTING MECHANISMS. THE BEST PERFORMANCE IS IN BOLDFACE, AND THE SECOND BEST IS UNDERLINED

Method	Large-vehicle		SHA	
	MAE	MSE	MAE	MSE
SE [49]	14.17	23.9	<u>50.85</u>	<u>83.95</u>
HDM [50]	<u>14.01</u>	<u>23.7</u>	52.91	86.89
PCSA [51]	14.19	24.19	53.37	86.08
Ours	**12.56**	**22.27**	**49.53**	**77.05**

TABLE XII

ABLATION STUDY ON BACKBONES. THE BEST PERFORMANCE IS IN BOLDFACE, AND THE SECOND BEST IS UNDERLINED

Method	Large-vehicle		SHA	
	MAE	MSE	MAE	MSE
ResNet-34 [52]	31.29	55.87	63.18	116.13
Twins-SVT [53]	<u>18.66</u>	55.88	<u>54.92</u>	<u>92.2</u>
MAE [54]	37.56	50.73	150.83	229.25
CVT-21 [55]	30.86	<u>44.61</u>	77.72	136.57
Ours(Vmamba-S)	**12.56**	**22.27**	**49.53**	**77.05**

in determining the final counting performance. As shown in Table X, the best performance is achieved when $K = 2$. When $K = 1$ or when too many rules are introduced ($K > 2$), training becomes unstable, resulting in a noticeable decline in counting performance.

Finally, to verify the advantages of fuzzy logic, we compare our method with other weighting mechanisms, SE [49], HDM [50], and PCSA [51], as shown in Table XI. Specifically, SE employs adaptive learning through squeeze and excitation to assess the importance of each channel. HDM generates the channel attention by leveraging interscale feature differences to capture more discriminative information. PCSA introduces an input-aware self-attention to calibrate channel features. In contrast, our method delivers markedly superior performance. This improvement primarily stems from the explicit fuzzy logic-based guidance, which encourages more balanced learning across regions with different response strengths, thereby enhancing the model's discriminative capability in complex scenarios.

4) Ablation Studies on Backbones: To evaluate the effect of different backbones on model performance, we conduct comparative experiments using several representative architectures, as summarized in Table XII. The selected backbones include ResNet-34 [52], Twins-SVT [53], MAE [54], CVT-21 [55], and VMamba-S [33]. All models are initialized from publicly available pretrained weights and fine-tuned for the remote sensing counting task. The results show that VMamba serves as a more effective feature extractor, owing to its ability to simultaneously model global context and local details, thus improving the representation of small targets in remote sensing scenarios.

5) Ablation Studies on Object Densities: To assess the adaptability of our method under varying density conditions, we conduct experiments on RSOC. The large-vehicle and small-vehicle subsets exhibit distinct density characteristics, averaging 96 and 531 targets per image, respectively, corresponding to low- and high-density scenarios. For detailed analysis, the test samples of each subset are further divided into low- and high-density levels using the mean values as a threshold. As shown in Fig. 4, our method achieves consistent and notable performance gains across both subsets, demonstrating strong robustness to different density scenarios.

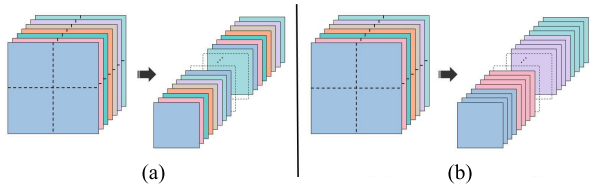


Fig. 3. Visual comparison of stacking methods for feature block partitioning. (a) Block-channel stacking. (b) Channel-block stacking.

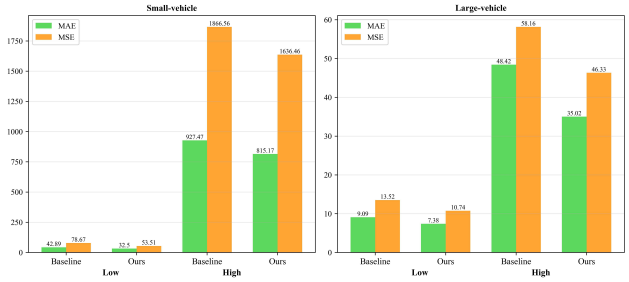


Fig. 4. Ablation studies on object densities.

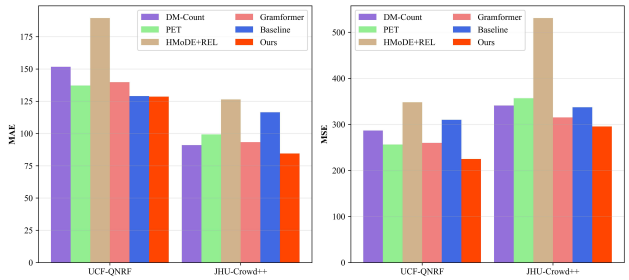


Fig. 5. Performance comparison of our method with other methods under cross-dataset evaluation.

D. Generalization Analysis

To evaluate the generalization capability of our model, we conduct cross-dataset validation. Due to the limited availability of remote sensing datasets, we perform validation on crowd counting datasets. Specifically, the model trained on SHA is tested directly on UCF-QNRF [56] and JHU-Crowd++ [57]. We compare our method with DM-Count [36], PET [40], HMoDE+REL [58], and Gramformer [12]. As shown in Fig. 5, the results demonstrate that our method significantly outperforms other methods. We observe that, compared with the baseline, our method achieves slightly better MAE on UCF-QNRF while yielding a substantially lower mse. This indicates improved stability when handling diverse and complex scenarios. On JHU-Crowd++, our method delivers

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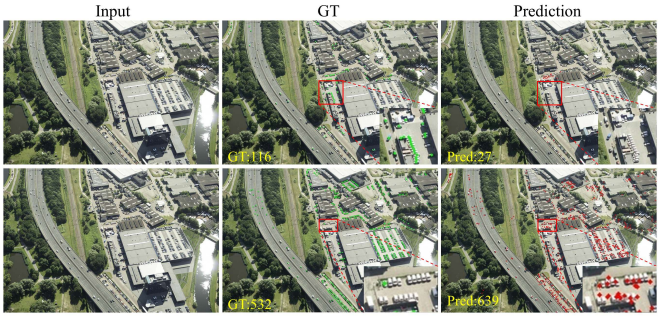


Fig. 6. Some failure cases of our method.

significantly stronger performance overall, further demonstrating its advantages.

E. Limitations and Discussion

Although our method achieves strong overall performance, it may still yield overestimated or underestimated results when distinguishing visually similar categories. Fig. 6 illustrates two failure cases, both captured from the same scene. The first row, taken from the large-vehicle dataset, is used for counting large vehicles, whereas the second row, taken from the small-vehicle dataset, is intended for counting small vehicles. Owing to the high-altitude perspective of remote sensing imagery, large vehicles often appear relatively small, causing the model to misclassify some large vehicles as small ones. This results in an underestimation of large vehicle counts and an overestimation of small vehicle counts (as indicated by the red bounding boxes). To overcome this limitation, future work could improve the model's capability to distinguish between targets that share similar semantics but belong to different categories.

V. Conclusion

This article addresses the generalization challenges posed by limited training data in remote sensing counting. To overcome this issue, we propose a novel counting network framework that integrates two key modules: a GdIR and a DRaM. The former captures the relationships among input features to construct a graph structure, which guides the regulator in adjusting the input distribution. This allows the network to prioritize salient information while mitigating the risk of overfitting. The latter employs fuzzy logic to identify and highlight discriminative regions, enabling the model to focus more effectively on critical local information. Extensive experiments conducted on multiple datasets confirm the effectiveness of the proposed approach, which achieves state-of-the-art performance. This work presents a promising solution for remote sensing counting in scenarios with limited annotated data.

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